PROSPECT 26-001

Proxmox SDN/BGP Integration

Submission plan for controlled discovery, design, implementation, validation, and handoff.

Estimated effort: 13-29 hours

Typical timeline: 2-4 working days

Core stance

Treat this as production network engineering: discovery and approval first, then staged change, validation, rollback, and documentation.

1. **Discovery**
   Topology, access, versions, route scope
2. **Design**
   SDN/BGP architecture and rollback
3. **Implement**
   Proxmox SDN, BGP/EVPN, route filters
4. **Validate**
   Peers, prefixes, workloads, logs
5. **Handoff**
   Runbook, diagram, acceptance evidence

DELIVERY STANCE

Why this should be handled as a controlled network change

The value is not only making BGP come up. The value is avoiding route leaks, access loss, and undocumented production drift.

Phase 1

Discover before changing

- Confirm Proxmox versions, node count, uplinks, bridges, MTU, and current SDN state.
- Inventory router/firewall platform, existing FRR or manual routing config, and access path.

Phase 2

Design the route boundary

- Define ASN plan, peer addresses, route targets, VNIs, exit nodes, and accepted prefixes.
- Use explicit route filters and rollback steps before production changes.

Phases 3-5

Validate evidence, not hope

- Verify BGP sessions, advertised and received prefixes, guest connectivity, failover behavior, and logs.
- Deliver final runbook, topology diagram, config notes, and acceptance criteria.

CLARIFICATION QUESTIONS

The client discussion should resolve these decision areas first

These map directly to section 1 of the submission plan and give us clean references for follow-up.

1.1-1.3 Environment

- Proxmox version and cluster shape
- Node count and interface roles
- Plain BGP, EVPN/VXLAN, or both

1.4-1.8 Routing Design

- Peer router/firewall platform
- Private or public ASN/prefix scope
- Exit node and active/active behavior

1.9-1.12 Existing Controls

- Current FRR/router configs to preserve
- Firewall/security policy boundaries
- Maintenance window and monitoring

1.13-1.14 Delivery Model

- VPN, bastion, and console access
- Configuration-only or full runbook/diagram handoff

Coverage across the technical surfaces that can break production

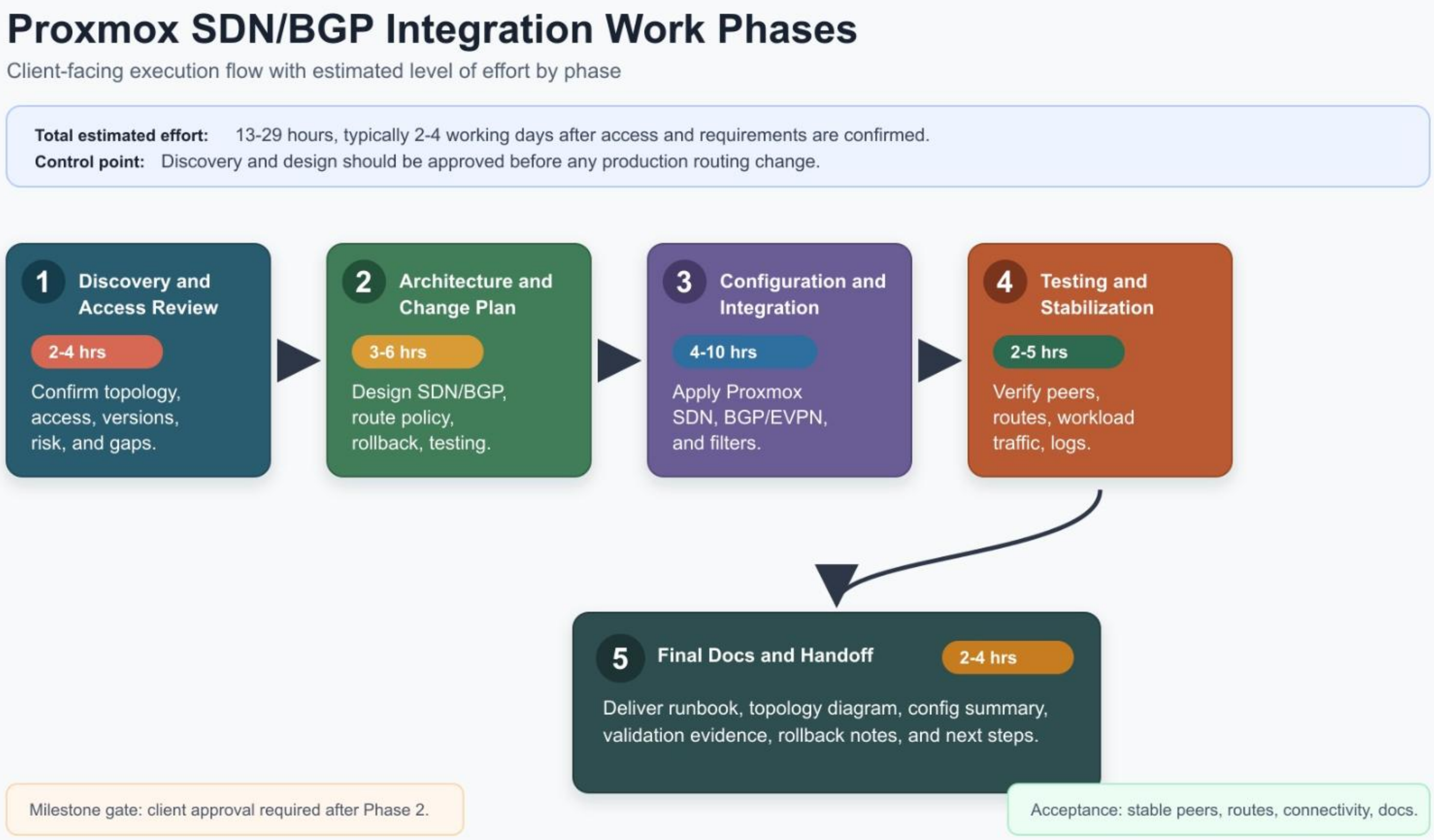
The plan covers both Proxmox SDN mechanics and the routing/security controls around them.

Plan Ref	Coverage Area	What We Complete	Client Evidence
4.1	**Proxmox SDN Review**	Cluster health, Linux bridges, bonds, VLAN-aware bridges, MTU, zones, VNets, subnets, IPAM.	Current-state notes and readiness findings.
4.2	**BGP/EVPN Design**	ASNs, router IDs, peers, route targets, VNIs, import/export behavior, failover model.	Approved target design and route plan.
4.3	**Security Controls**	Management isolation, prefix filters, peer policy, firewall boundary review, public route guardrails.	Explicit route/security policy decisions.
4.4-4.5	**Implement + Validate**	Backups, staged config, upstream coordination, BGP state, prefixes, workload traffic, logs.	Validation checklist and test results.
4.6	**Documentation**	Final-state diagram, key configs, validation commands, rollback, maintenance notes.	Runbook and handoff package.

WORK PHASES

Execution flow with level of effort

The flow chart turns the proposal into a simple client-call discussion path.



Reference: Work phases from section 5, timeline from section 6.

PHASE DETAIL

Level of effort is concentrated where routing risk is highest

Configuration effort varies by cluster size, router complexity, and whether the client needs plain BGP or EVPN/VXLAN.

Ref	Phase	Effort	Primary Output
5.1	**Discovery and Access Review**	2-4 hrs	Access, topology, versions, risk notes, missing information
5.2	**Architecture and Change Plan**	3-6 hrs	Target design, route policy, change sequence, rollback, validation plan
5.3	**Configuration and Integration**	4-10 hrs	Proxmox SDN, BGP/EVPN peers, upstream coordination, route filters
5.4	**Testing and Stabilization**	2-5 hrs	Peer state, route advertisements, workload traffic, failover/log checks
5.5	**Final Documentation and Handoff**	2-4 hrs	Runbook, topology diagram, config summary, validation evidence

Total estimated effort 13-29 hours

Phase gate Client approval required after design before production changes.

RISK CONTROLS

The plan is built to prevent the common failure modes

These risks are not theoretical in routing work, so the proposal makes them explicit before implementation.

9.1 Route leaks

- Explicit prefix filters, peer policy, and pre-approved route lists.

9.2 Access loss

- Console or out-of-band path, staged changes, and rollback commands.

9.3 MTU mismatch

- Review physical, bridge, VNet, VXLAN, and guest MTU before rollout.

9.4 Config conflict

- Decide whether Proxmox SDN or manual FRR owns each routing function.

9.5 Asymmetric traffic

- Define exit-node policy, route preference, and test traffic paths.

Proposal guardrail: do not accept production implementation without discovery, route scope, access model, rollback, and maintenance window.

A clean close gives the client a way to approve the work

The strongest submission is a bounded project with objective acceptance criteria.

11.1

Discovery and Design

Current-state review, target architecture, implementation plan, rollback plan.

11.2

Implementation and Validation

Configured SDN/BGP integration, route exchange, connectivity validation.

11.3

Documentation and Handoff

Runbook, topology diagram, config summary, maintenance instructions.

Acceptance criteria

- Stable BGP peers
- Approved SDN prefixes advertised
- Unapproved prefixes blocked
- VM/LXC connectivity confirmed
- Management/storage/migration isolated
- Runbook delivered

Recommended next action: send the clarification questions first, then price discovery/design as Milestone 1 before committing to production implementation.